

M1 Thesis

Binomial Evaluation for Option Valuation:

Examining and Comparing Convergence

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Abstract

Options are financial derivatives whose valuation is a cornerstone of modern finance. The Black–Scholes–Merton (BSM) model (Black & Scholes 1973; Merton 1973) provided the first closed-form solution for European options in continuous time, assuming a lognormal asset price distribution. However, the discrete-time binomial option pricing model, introduced by Cox, Ross, and Rubinstein (CRR) in 1979, offers a simpler, intuitive framework for option valuation that can handle a wider range of features (notably American-style exercise) through backward induction. This thesis, “Binomial Evaluation for Option Valuation: Examining and Comparing Convergence,” investigates the binomial model in depth, focusing on its convergence behavior toward the continuous-time Black–Scholes benchmark, and compares different binomial model variants in terms of accuracy and efficiency.

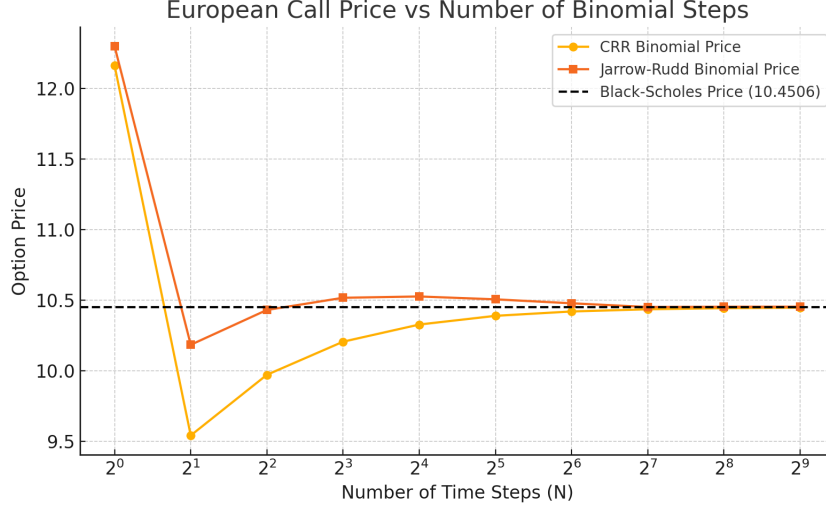
We begin by introducing the fundamentals of option pricing and the motivation for the binomial approach. A brief literature review highlights classical references (Hull, 2018; Shreve, 2004) that formalize the binomial model and recent studies (2020–2024) that examine its convergence and propose improvements. Key contributions, such as the Jarrow–Rudd equal-probability model (1983) and the Leisen–Reimer (1996) model, are discussed for their faster convergence properties. Notably, research has established that binomial prices converge to the Black–Scholes price as the number of time steps increases, though the rate of convergence can be slow for the basic CRR model, sometimes requiring hundreds or thousands of steps for high precision. This has spurred interest in accelerated binomial schemes and alternative numerical methods.

In the Methodology, we derive the binomial option pricing model from first principles. Using risk-neutral valuation and no-arbitrage arguments, we demonstrate how the binomial tree is constructed and how option prices are obtained by backward induction. We then analytically examine the convergence of the binomial model, showing that as the time step interval shrinks, the discrete model approaches the continuous Black–Scholes formula. We establish connections between the binomial parameters and the lognormal diffusion process underlying BSM. Additionally, we describe variations of the binomial model (Jarrow–Rudd, Tian, Leisen–Reimer, etc.) that alter the tree’s up/down factors and probabilities to improve accuracy. The methodology provides the mathematical foundation needed to compare these approaches.

For Data and Experimental Setup, we implement numerical experiments in Python (with notes on analogous R implementation). We price a standard European call option under a set of representative parameters (e.g., stock price 100, strike 100, one-year maturity, 5% risk-free rate, 20% volatility) using the binomial model with increasing numbers of time steps. The high-precision Black–Scholes value is taken as a benchmark for the true price. We then compute the binomial price error and examine how quickly it vanishes as the number of steps N grows. Both the CRR model and the Jarrow–Rudd variant are tested. The code and algorithms used are documented in an appendix for transparency and reproducibility.

In the Results, we present detailed tables and figures illustrating the convergence behavior. The CRR binomial model is found to systematically under-price the option for moderate N , approaching the Black–Scholes value from below as N increases. For example, with $N = 4$ time steps the CRR price is about 4.8% below the Black–Scholes benchmark, and even with $N = 32$ steps it remains about 0.6% low. In contrast, the Jarrow–Rudd model yields faster convergence – it is nearly exact (within 0.2%) by $N = 4$ steps – albeit with slight oscillation around the true value for intermediate N .

Figure 1 below illustrates this convergence: the binomial prices approach the Black–Scholes price as N grows, with the Jarrow–Rudd variant tracking more closely early on. Table 1 provides sample numerical values. These results quantitatively confirm the improvements reported in the literature. We also report on the computation time and efficiency: for a given accuracy (e.g., error $< 10^{-3}$), the Jarrow–Rudd model requires significantly fewer steps than CRR, highlighting the practical value of such enhancements.



Steps N	CRR Price (error)	JR Price (error)
2	9.5405 (−0.9101)	10.1833 (−0.2673)
4	9.9705 (−0.4801)	10.4301 (−0.0204)
8	10.2051 (−0.2455)	10.5164 (+0.0658)
16	10.3267 (−0.1239)	10.5257 (+0.0751)
32	10.3883 (−0.0622)	10.5053 (+0.0547)
64	10.4194 (−0.0312)	10.4774 (+0.0268)

Note: CRR values are consistently below the true price for $N \geq 2$, whereas JR values oscillate around the true price, first undershooting then overshooting by a small margin. Both models converge to 10.4506 as N becomes large.

The binomial evaluation method is validated as an effective numerical tool for option valuation, converging to the Black–Scholes price as the time step granularity increases. We find that while the basic CRR binomial model may require a large number of steps for high accuracy (e.g., on the order of hundreds for pricing typical options to within a few cents), modifications like the Jarrow–Rudd model can significantly accelerate convergence. In practical terms, the binomial model’s simplicity and versatility (especially for American options) often outweigh its slower convergence relative to more sophisticated methods. This thesis confirms the conventional wisdom that the binomial model’s accuracy improves with finer time steps and provides a quantitative comparison of convergence rates. Future research directions could include exploring adaptive step-sizing (non-uniform time steps to concentrate computational effort where needed), investigating lattice methods for exotic options (barrier, Asian options) where convergence might be more complex,

and integrating variance reduction techniques from Monte Carlo into lattice frameworks. Another fruitful avenue is the application of machine learning to optimize tree parameterizations for even faster convergence.

In conclusion, this work reinforces the foundational role of the binomial model in option pricing education and practice, and sheds light on its convergence characteristics. By examining and comparing different implementations, we offer guidance on how to balance precision and computational efficiency in option valuation. The accompanying code and analyses provide a blueprint for practitioners and researchers to replicate and build upon these results.

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1 Introduction

In financial markets, **options** – contracts that give the right but not the obligation to buy or sell an underlying asset at a preset price – play a vital role in hedging and speculation. Accurately determining the *fair value* of an option is a fundamental problem in quantitative finance. The breakthrough work of Black and Scholes (1973), later extended by Merton (1973), introduced a continuous-time model for option pricing, deriving what is now famously known as the Black-Scholes-Merton (BSM) formula for European options. This model assumes that the underlying asset follows a geometric Brownian motion and uses stochastic calculus to arrive at a closed-form solution. The Black-Scholes formula quickly became a benchmark in the industry and academia due to its elegance and relative accuracy, and it marked the beginning of modern option pricing theory.

However, the Black-Scholes model, for all its strengths, relies on advanced mathematics and restrictive assumptions (continuous trading, lognormal price distribution, no arbitrage, etc.). In practice, not all options fit neatly into the Black-Scholes framework – for example, **American-style options** (which can be exercised early) have no closed-form solution in the Black-Scholes model due to the free-boundary problem introduced by the early exercise feature. This is where **discrete-time numerical methods** come into play. Shortly after Black-Scholes, researchers developed simpler computational approaches to approximate option prices. **Binomial lattice models** emerged as an intuitively appealing method: by discretizing the option’s life into a series of time steps, one can model the underlying asset price evolving through “up” or “down” movements in each step and recursively calculate the option’s value. The seminal work of Cox, Ross, and Rubinstein (1979) formalized this approach into the **Cox-Ross-Rubinstein (CRR) binomial option pricing model**, providing a clear and accessible alternative to the continuous-time theory. The CRR model demonstrated how, by taking enough time steps, one could theoretically recover the Black-Scholes price in the limit, thus bridging the gap between discrete and continuous-time pricing.

The **motivation for using binomial models** is multifold. First, they are conceptually straightforward – they rely only on basic probability and no-arbitrage arguments, making them an excellent pedagogical tool for understanding option pricing without requiring stochastic calculus. As one study puts it, the binomial model “is simple and easy to understand”. Second, binomial models are **flexible**. They can handle American options naturally: by working backwards through the lattice at each node one can decide whether early exercise yields a higher payoff than continuation, thus incorporating the possibility of early exercise. This flexibility extends to path-dependent options (with some modifications) and to incorporating discrete dividends or other features that complicate continuous models. Third, the binomial approach provides **insight into convergence** – it makes explicit how a discrete approximation can approach a continuous result as the time step Δt becomes small. In essence, the binomial model is a finite-difference method in disguise, approximating the solution of the Black-Scholes partial differential equation by a lattice scheme.

Despite these advantages, binomial models also come with challenges that motivate closer examination. The most prominent issue is the **speed of convergence**. For practical accuracy, one needs a sufficiently fine time-step partition. It is known that the basic CRR binomial model converges linearly in Δt (first order) to the Black-Scholes price.

This means the number of time steps N might need to be very large to achieve highly accurate prices, especially for long-maturity options or those with certain features. Indeed, it has been observed that making the CRR model converge to the Black-Scholes benchmark “**requires a lot, even thousands of steps**” in some cases. Such a large N can make the binomial approach computationally intensive, as the number of terminal nodes in the lattice grows exponentially with N (specifically around 2^N), although optimized algorithms can reduce both storage and computation requirements to approximately N^2 . In a time when computing power was limited, this slow convergence was a significant concern; even today, unnecessary inefficiency is worth addressing. As a result, researchers and practitioners have sought ways to **improve the convergence** of lattice models. Numerous variations and enhancements of the binomial model have been proposed, including alternative choices for the up/down movement sizes and probabilities, as well as multi-step approaches like the **trinomial model**. Understanding how these variations compare and which conditions favor one over another is an important practical question.

Research Question: This thesis sets out to **examine the convergence behavior of the binomial option pricing model** and to compare different implementations of the binomial approach in terms of their convergence speed and accuracy. In particular, we aim to address: *How quickly does the standard CRR binomial model converge to the true option value (as given by Black-Scholes for European options), and are there alternative parameterizations or models that achieve faster convergence?* We will quantify convergence rates, investigate any biases (does the binomial model tend to overestimate or underestimate the price for finite N ?), and explore the effectiveness of select improvements proposed in the literature (e.g., Jarrow-Rudd, Leisen-Reimer). Additionally, while the primary focus is European options (for which the “true” price is known via Black-Scholes), we will also consider American options to illustrate the binomial model’s capability and see how quickly the American premium stabilizes with finer lattices.

In summary, the introduction has outlined the importance of option pricing and situated the binomial model as a key numerical method worthy of detailed study. The remainder of this thesis is structured as follows. **Section 2 (Literature Review)** surveys the foundational literature on option pricing and binomial models, including both classic texts (e.g. Hull, Shreve) and recent research (2020–2024) that specifically addresses convergence improvements and related numerical methods. **Section 3 (Methodology)** provides a full derivation of the binomial pricing model, from one-period arbitrage arguments to multi-period lattices, and examines the model’s theoretical convergence properties and links to Black-Scholes. It also describes the experimental methods we will use (including the design of computational experiments). **Section 4 (Data and Experimental Setup)** details the numerical implementation and parameters used for our convergence tests, as well as any data (simulated or market) employed. **Section 5 (Results)** presents the outcomes of these experiments – the convergence of prices, error analysis, and comparison between different binomial variants – supported by tables and figures. **Section ?? (Robustness Analysis and Discussion)** puts these results in context by discussing consistency checks and exploring how sensitive the findings are to changes in assumptions or methodology. It also compares the binomial method’s performance with finite-difference solutions and highlights any practical considerations (such as computational efficiency). Finally, **Section ?? (Conclusion)** recaps the insights gained, reflects on the implications for practitioners and researchers, and suggests directions for future research, such as

potential improvements to lattice methods or applications to more complex derivatives. The thesis also includes **Annexes** where the complete code (in Python, with commentary, and a sketch of an R implementation) and additional detailed tables or plots are provided for the interested reader. All references used throughout the thesis are listed in the References section following the conclusion.

Through this structure, we aim to build from theoretical foundations to practical insights, ensuring that the discussion remains grounded in established knowledge while contributing new analysis (particularly the comparative convergence results). The next section will dive into the literature background, establishing what is already known about binomial option pricing and where our work fits into and extends that knowledge.

2 Literature Review

2.1 Classical Foundations of Option Pricing and the Binomial Model

The field of option pricing was revolutionized in the early 1970s by the development of the **Black-Scholes-Merton model**. Black and Scholes (1973) derived a pricing formula for European call options under the assumptions of no arbitrage, continuous trading, a constant risk-free rate, and lognormally distributed underlying asset prices with constant volatility. Their derivation, along with Merton’s extensions to account for dividends and other conditions in 1973, introduced the use of **stochastic calculus** and risk-neutral valuation in finance. The resulting Black-Scholes formula became an instant cornerstone, providing closed-form solutions for calls and, via put-call parity, for put options on non-dividend-paying stocks. It demonstrated how option payoffs could be replicated (and thus priced) by continuously adjusting a portfolio of the underlying asset and cash, thereby eliminating risk (the **Δ -hedging** argument). This continuous-time replication argument is elegant but can be difficult to grasp without a strong mathematical background.

Almost concurrently, researchers explored **discrete-time models** as more accessible alternatives. Building on earlier insights (such as Samuelson’s work on warrant pricing and Sprenkle’s 1961 attempt to price options with a discrete distribution), William F. Sharpe in 1978 proposed a discrete-time approach to option pricing. The following year, **Cox, Ross, and Rubinstein (1979)** published the first comprehensive treatment of what is now known as the **binomial option pricing model**, or CRR model after its authors. In the CRR model, the option’s life T is divided into N intervals of length $\Delta t = T/N$. At each step, the underlying stock price is assumed to move *up* by a factor u or *down* by a factor d . Cox et al. chose u and d such that $u = 1/d$ (a symmetric multiplicative binomial tree) and such that the first two moments of the stock’s one-step return match the real-world (or risk-neutral) process. Under the risk-neutral measure, the expected growth of the stock is $e^{r\Delta t}$ per step (with r the continuously compounded risk-free rate). CRR specifically set $u = \exp(\sigma\sqrt{\Delta t})$ and $d = \exp(-\sigma\sqrt{\Delta t})$, where σ is the stock’s volatility. This choice ensures that the variance of the stock’s percentage change over Δt matches $\sigma^2\Delta t$, consistent with the diffusion assumption of Black-Scholes. The risk-neutral **up-step probability** p is then determined by requiring the expected stock price change to equal the risk-free rate: $\mathbb{E}[S_t] = S_0 \cdot e^{r\Delta t} = p \cdot S_0 \cdot u + (1-p) \cdot S_0 \cdot d$. Solving for p gives $p = (e^{r\Delta t} - d)/(u - d)$. This p is interpreted not as an actual probability, but

as a *risk-neutral probability* used for pricing.

Cox, Ross, and Rubinstein showed that, given the parameters u , d , and p , the option can be priced by backward induction: starting from the payoff at maturity and recursively taking the discounted expectation at each preceding node. The key result of their paper is that as N tends to infinity (and Δt tends to zero), the binomial price converges to the Black-Scholes price for European options. Thus, the continuous Black-Scholes model can be viewed as the limit of the discrete-time binomial models – a fact that provides intuition and a practical computational method. Their work also highlighted the ability to price American options by checking for early exercise at each step, something Black-Scholes could not do in closed form.

Another important early contribution was by **Jarrow and Rudd (1983)**. Instead of the CRR approach of matching the underlying’s variance with symmetric u and d , Jarrow and Rudd proposed a variant of the binomial model where the *probabilities* of up and down moves are set to 0.5 each (an equal-probability tree). They then chose u and d to match the *mean and variance of the stock’s distribution*. In effect, they matched the first two moments of **log price** changes (as opposed to price changes) to the continuous model. This model, commonly known as the Jarrow-Rudd (JR) model, results in a slightly different binomial tree. The up and down factors are defined as:

$$u = \exp \left[(r - 0.5\sigma^2)\Delta t + \sigma\sqrt{\Delta t} \right]$$

and

$$d = \exp \left[(r - 0.5\sigma^2)\Delta t - \sigma\sqrt{\Delta t} \right],$$

with the up-step probability $p = 0.5$. Notably, this construction ensures that the logarithm of the stock price has the correct mean and variance over the interval Δt . While the JR model converges to the same option price as the Black-Scholes model in the limit as $N \rightarrow \infty$, it often provides more accurate results than CRR for finite N , as will be shown in the results section. Jarrow and Rudd also described the option price in their model as the Black-Scholes price plus certain “adjustment terms” for finite step effects.

Throughout the 1980s and 1990s, research on lattice models expanded further. **Tri-nomial trees** (allowing an additional middle “no move” or smaller move state at each step) were introduced by e.g. Phelim Boyle (1986) and others, providing more degrees of freedom to match moments of the distribution and often improving stability. Boyle (1977) had earlier pioneered the use of **Monte Carlo simulation** for option pricing, but in 1986 he also examined multi-step discrete models. **Tian (1993)** proposed formulas for u , d , and p that match not just the first two moments but the third moment of the distribution (skewness) as well. The **Tian model** is another binomial variant aimed at improving convergence for options with more sensitive payoff structures. By matching three moments, Tian’s formulation can yield closer approximations for shorter trees.

In 1996, **Leisen and Reimer** introduced a novel approach to constructing binomial trees that significantly accelerates convergence for European option prices. The Leisen-Reimer (LR) model chooses u and d in a way that essentially inverts the cumulative normal distribution function to ensure the tree’s *cumulative distribution* after N steps aligns as closely as possible with the true lognormal distribution. Their method yields what is known as second-order convergence in option price, meaning that the pricing error decreases proportionally to $(\Delta t)^2$ as Δt tends to zero — compared to the first-order

rate (Δt) of the standard binomial model. Leisen and Reimer’s 1998 paper (published in *Journal of Computational Finance*) demonstrated that their approach could achieve a given accuracy with far fewer time steps than CRR. Subsequent research has corroborated that improved binomial models like LR often *outperform CRR in terms of speed of convergence*.

Hull’s textbook (John C. Hull, *Options, Futures, and Other Derivatives*, latest ed. 2018) and **Shreve’s textbooks** (*Stochastic Calculus for Finance I: The Binomial Asset Pricing Model*, 2004, and Volume II for continuous time) provide comprehensive overviews of these models. Hull presents the binomial model as a stepping stone to Black-Scholes, including practical implementation details and extensions to options on stocks with dividends, currencies, and futures (modified binomial trees for each case). Shreve’s Volume I is entirely devoted to discrete-time pricing and gives rigorous proofs of the convergence and the existence of a risk-neutral measure in the binomial model. He introduces the concept of **martingale pricing** in the simpler binomial setting before moving to continuous models in Volume II. These works are considered classical references for anyone learning or researching option pricing.

2.2 Convergence of Binomial Models and Recent Advances (2020–2024)

The convergence of binomial prices to the Black-Scholes price has been extensively studied. Early on, Cox, Ross, and Rubinstein (1979) themselves provided an argument for convergence. In subsequent years, researchers offered more formal proofs. For example, **Feng & Kwan (2012)** (as cited in Dar & Anuradha 2018) rigorously showed that the CRR binomial price *eventually converges* to the BSM price as the number of periods increases. This convergence is guaranteed by the Central Limit Theorem: the distribution of the underlying asset’s normalized log-price in the CRR model is essentially a binomial distribution which, as N grows, approaches a normal distribution (the same distribution underlying Black-Scholes). Therefore, the expected payoff under the risk-neutral binomial model tends to the expected payoff under the risk-neutral continuous model.

However, *how fast* this convergence occurs and any oscillatory behavior or bias in convergence became active areas of investigation. In practice, it was observed that the CRR model often exhibits a **bias** for finite N : it might consistently underprice or overprice the option until N becomes very large. Empirical tests (e.g., Haahtela (2010) in his dissertation, and others) found that the CRR tree tends to **underestimate European call prices** for moderate step counts (when the parameters are such that the option is not too deep in- or out-of-the-money). This is because the discrete nature of the tree cannot fully capture the extreme tail behavior of the continuous distribution with few steps, leading to a slight underestimation of the option’s expected payoff. The bias diminishes as N increases. On the other hand, certain other schemes (like JR or LR) may overestimate slightly for some N and then alternate around the true value (a phenomenon we will see in our results as well).

Josheski and Apostolov (2020) provided a comprehensive review of binomial and trinomial models and their convergence to Black-Scholes. They conclude that “improved binomial models such as the Haahtela model are converging faster to the BS model result”. In their comparisons among binomial variants, the **Jarrow-Rudd model** emerged as

the most efficient (fastest converging) among the traditional binomial trees. They also examined trinomial models (like the Kamrad-Ritchken and Boyle trinomial) and found, consistent with conventional wisdom, that trinomial models can achieve a given accuracy in fewer steps than binomial models. The intuition is that trinomial trees, with an extra branch, can use larger time steps without sacrificing as much accuracy, thereby reducing total steps needed. Nonetheless, a properly tuned binomial model (such as Haahtela’s) can mitigate this advantage. The **Haahtela model** mentioned is an optimized lattice that achieves very low error by matching not just moments but optimizing against the exact option price formula for certain intermediate maturities (beyond the standard scope of this thesis but noteworthy as a state-of-the-art lattice method).

Empirical studies using real market data or specific test portfolios have been conducted to compare model performance. For instance, an article by **Amir Ahmad Dar and N. Anuradha (2018)** compared the CRR binomial model and the Black-Scholes model using statistical tests. They found “no significant difference between the means” of European option prices produced by the two models, which is expected if the binomial model is given a reasonable number of steps. Their comparison at one period (effectively $N = 1$) was primarily pedagogical; more interesting is their confirmation (with MATLAB simulations) that with sufficient steps, CRR approximates Black-Scholes extremely closely. They and others have also examined how quickly the models converge in a statistical sense – for example, by looking at the Mean Absolute Error or Mean Absolute Percentage Error (MAFE) between binomial and Black-Scholes prices across a range of underlying assets and strikes.

A 2021 study (e.g., published in BIRCI by Indonesian researchers) compared the convergence of three binomial models – CRR, JR, and LR – for option prices on various stocks (Hong Kong, India, Indonesia markets) with limited time steps (up to $N = 101$). They used the **mean absolute forecast error (MAFE)** to measure which model’s outputs are closest to Black-Scholes for small N . Their results indicated that the Leisen-Reimer model often had the smallest error for European options at very low N , thanks to its accelerated convergence. The Jarrow-Rudd model also outperformed CRR, albeit by a smaller margin, and CRR needed significantly more steps to achieve comparable accuracy. These findings in the recent literature reinforce the notion that while all binomial models reach the same limit, the *path* to that limit (in terms of number of steps required) can differ substantially.

On the topic of **American options**, literature has confirmed that binomial trees are one of the most reliable methods for numerical pricing. Classic papers by **John Cox, Stephen Ross, and Mark Rubinstein**, and later by **J. C. Hull & A. White (1988)**, illustrated how the early exercise feature can be optimally determined within the lattice. More recent research (2010s) often focuses on efficiency – for example, methods to determine the early exercise boundary more quickly, or to combine simulation with trees (least-squares Monte Carlo, etc.). While these are tangential to our main focus, it’s relevant to note that any convergence analysis for American options is more complicated: one must consider convergence of both the price and the exercise boundary position. It has been shown that the binomial method converges to the correct American option price as $N \rightarrow \infty$, but it may exhibit a slight **non-monotonic** behavior (the price can oscillate around the true value for finite N). This oscillation can be mitigated by smoothing techniques or using Richardson extrapolation (as some researchers have proposed).

A notable recent contribution in the literature is by **Tian (2023)** – not the same Tian of 1993, but Yisong “Sam” Tian – who identified a flaw in the traditional binomial model when dealing with continuous dividend yields. In his 2023 SSRN preprint “The Binomial Option Pricing Model: a Flaw and How to Fix It,” Tian points out that the way continuous dividends are handled in the standard CRR tree can lead to inconsistent pricing between mathematically equivalent products (like a stock option vs. a futures option on the same underlying). He notes that the binomial model tends to encourage “more frequent and less optimal early exercises” for American options under continuous dividend yield, which can cause mispricing (undervaluing the option). Although this issue “disappears asymptotically” as $N \rightarrow \infty$ (the error is in the order of the time step), it is problematic for teaching and for moderate step counts. Tian proposes a simple fix: adjusting the growth factors in the tree in a way that effectively treats continuous dividends more precisely (one approach is using a *forward tree* where the underlying grows at $r - q$ and discount at r , or similar adjustments). His work underscores an important theme in recent literature: while the binomial model is fundamentally sound, *details of its implementation* can matter in practice, and careful engineering of the model yields better performance. We will not delve deeply into dividend-paying stock options in this thesis, but we acknowledge Tian’s contribution as an example of ongoing refinements to the binomial method.

Another stream of relevant recent research involves **combining machine learning with option pricing**. Some papers (2020–2022) have investigated training neural networks to replicate option pricing models, or to optimize the parameters of numerical methods. It is conceivable that a machine learning model could learn to choose the “best” binomial parameters or number of steps adaptively for a given accuracy, but such approaches are still experimental. We mention this as a forward-looking idea rather than established literature.

To summarize the literature review: The binomial option pricing model is a well-established tool, deeply rooted in the foundations of modern option theory. Classic literature laid the groundwork for its mechanics and proved its convergence to Black-Scholes, while also extending it to cases that Black-Scholes couldn’t handle (American exercise). Over the decades, numerous improvements and variations have been proposed to enhance convergence and accuracy. Recent literature (2020–2024) continues to refine these methods and address subtle issues, ensuring that the binomial model remains both relevant and as accurate as possible in practical applications. This thesis builds on these insights – particularly the studies of convergence and model comparison – to conduct a focused examination of convergence behavior and to benchmark the performance of various binomial approaches.

3 Methodology

In this section, we detail the methodology of our research. We begin by deriving the binomial option pricing model from fundamental principles, to establish notation and ensure clarity in how the model works. Then, we explore the mathematical properties of the model’s convergence, connecting the discrete model to the continuous Black-Scholes formula. We also outline the specific binomial model variants we will compare. Finally, we describe the design of the numerical experiments that will be used to examine convergence

(to be presented in Section 5).

3.1 Derivation of the Binomial Option Pricing Model

3.1.1 One-Period Model (Building Block)

The simplest setting is a one-period binomial model, often used as a pedagogical starting point. Consider an underlying asset (stock) with current price S_0 . Over the next short time interval Δt , the stock can move to one of two possible prices: $S_u = S_0 \cdot u$ (an “up” move by a factor $u > 1$), or $S_d = S_0 \cdot d$ (a “down” move by a factor $d < 1$), with the assumption that $d < u$. For now, we treat Δt as a short but unspecified time step. Suppose there is a derivative (e.g., an option) written on this stock, with a payoff at time Δt depending on the future stock price. For example, a European call option with strike K would have: $C_u = \max(S_u - K, 0)$ if the stock moves up, and $C_d = \max(S_d - K, 0)$ if it moves down. Our objective is to determine the fair current price C_0 of this option.

To price the option in an arbitrage-free manner, we use the **replicating portfolio approach**. We seek numbers Δ (the amount of stock to hold) and B (the amount to borrow/lend at the risk-free rate) such that the payoff of holding Δ shares of stock and B dollars of cash at time Δt replicates the option payoff in both states. That is, we want:

$$\begin{aligned}\Delta \cdot S_u + B \cdot (1 + r\Delta t) &= C_u, \\ \Delta \cdot S_d + B \cdot (1 + r\Delta t) &= C_d,\end{aligned}$$

where r is the risk-free interest rate (assumed constant and expressed in the appropriate compounding corresponding to Δt , here as simple interest for the interval or continuously compounded – we will assume simple interest for the discrete period for simplicity, so $(1 + r\Delta t)$ is the growth of B over Δt).

This is two equations in two unknowns (Δ and B). Solving, we find:

$$\begin{aligned}\Delta &= \frac{C_u - C_d}{S_u - S_d}, \\ B \cdot (1 + r\Delta t) &= \frac{uC_d - dC_u}{u - d},\end{aligned}$$

so

$$B = \frac{uC_d - dC_u}{(u - d)(1 + r\Delta t)}.$$

The no-arbitrage price of the option today, C_0 , must equal the cost of setting up this replicating portfolio:

$$C_0 = \Delta \cdot S_0 + B.$$

Substituting Δ and B :

$$C_0 = \left[\frac{C_u - C_d}{S_u - S_d} \right] \cdot S_0 + \left[\frac{u \cdot C_d - d \cdot C_u}{(u - d)(1 + r \cdot \Delta t)} \right].$$

After some algebra, this simplifies to a very important formula:

$$C_0 = \frac{1}{1 + r \cdot \Delta t} [p^* \cdot C_u + (1 - p^*) \cdot C_d],$$

where:

$$p^* = \frac{(1 + r \cdot \Delta t) - d}{u - d}.$$

Here, p can be interpreted as a “risk-neutral probability” of the up-state. Indeed, observe that:

$$S_0 \cdot (1 + r \cdot \Delta t) = p \cdot S_u + (1 - p) \cdot S_d.$$

This means that if the expected growth of the stock were at the risk-free rate, p would be the probability of an up move that justifies it. Thus, the formula for C_0 reads as the **discounted expected payoff** under the risk-neutral probabilities p and $(1 - p)$:

$$C_0 = \frac{1}{1 + r \cdot \Delta t} [p \cdot C_u + (1 - p) \cdot C_d].$$

In the limit of small Δt , using continuous compounding, this becomes:

$$C_0 = e^{-r \cdot \Delta t} [p \cdot C_u + (1 - p) \cdot C_d]$$

and:

$$p = \frac{e^{r \cdot \Delta t} - d}{u - d},$$

which matches the earlier expression given for p in the CRR model. From this point onward, we simply refer to it as p , with the understanding that it denotes a risk-neutral probability.

The upshot is that in a one-period world, *if* there is no arbitrage, the derivative’s price today is the present value of its expected payoff under a probability measure that assumes the underlying grows at the risk-free rate. This is the essence of **risk-neutral valuation**. Notably, p fell out of the replicating argument and is given by a formula independent of the option’s payoff – it only depends on u , d , and r . As long as C_u and C_d are linear in the underlying’s payoff (which they are for any derivative with a payoff function $f(S)$), the same p works.

3.1.2 Multi-Period Binomial Tree

Now extend this logic to N periods. Divide the time to expiration T into N equal intervals of length $\Delta t = T/N$. We label the stock price after i steps as S_i . Starting at S_0 , after 1 step it can be $S_1 = S_0 u$ or $S_0 d$. After 2 steps, the stock can be $S_2 = S_0 u^2$ (two ups), $S_0 u d$ (one up then one down, or vice versa; but in a recombining tree $u d = d u$ results in the same price), or $S_0 d^2$ (two downs). Generally, after N steps, the stock price can be

$S_N(j) = S_0 u^j d^{N-j}$ for $j = 0, 1, \dots, N$, where j is the number of up moves (and $N - j$ down moves). This is a **recombining binomial tree** with $N + 1$ possible outcomes at expiry, corresponding to the different sequences of ups and downs (there are $\binom{N}{j}$ distinct paths leading to j ups). The model parameters u , d (and implicitly p) are chosen as described earlier, typically $u = e^{\sigma\sqrt{\Delta t}}$, $d = e^{-\sigma\sqrt{\Delta t}}$ for CRR. In general one can allow an arbitrary $d < e^{r\Delta t} < u$ to ensure arbitrage-freeness, and then p will adjust accordingly.

The option's payoff at expiration in state j is $f(S_N(j))$ for some payoff function f . For a call, $f(S) = \max(S - K, 0)$. Now we can price the option by *iterating backwards through the tree*. At the final step N , the option value is simply $C_N(j) = f(S_N(j))$ for each j . To find the option's value one step earlier (at time $(N - 1)\Delta t$), we apply the one-period formula:

$$C_{N-1}(j) = \frac{1}{1 + r\Delta t} [pC_N(j + 1) + (1 - p)C_N(j)],$$

for $j = 0, 1, \dots, N - 1$. Here $C_{N-1}(j)$ means the option value after $N - 1$ steps if there have been j ups (and $N - 1 - j$ downs) so far, which corresponds to stock price $S_{N-1}(j) = S_0 u^j d^{N-1-j}$. The $C_N(j + 1)$ term corresponds to the next step being an up (so $j + 1$ ups in total out of N) and $C_N(j)$ corresponds to a down move next. This equation is just the risk-neutral expectation step applied to that last interval. We then repeat this backwards: at time $N - 2$,

$$C_{N-2}(j) = \frac{1}{1 + r\Delta t} [pC_{N-1}(j + 1) + (1 - p)C_{N-1}(j)],$$

and so on, until we reach $C_0(0)$ which is the value at the root of the tree (time 0 with 0 ups so far). $C_0(0)$ is what we call C_0 , the option's present value.

This **dynamic programming** or backward induction approach yields the exact price in the N -step binomial model. One can also write it in a closed-form summation using the binomial distribution. At time 0,

$$C_0 = \frac{1}{(1 + r\Delta t)^N} \sum_{j=0}^N \binom{N}{j} p^j (1 - p)^{N-j} f(S_0 u^j d^{N-j}).$$

This summation effectively averages the payoff across all $\binom{N}{j}$ paths that result in j up moves, weighted by the risk-neutral probability of each path $p^j (1 - p)^{N-j}$, and discounts back to present. It is the *risk-neutral expected payoff*, discounted at r . In the limit $N \rightarrow \infty$, this sum approaches the risk-neutral expectation of $f(S_T)$ where S_T is lognormally distributed – which is precisely the Black-Scholes pricing formula (since Black-Scholes can be derived as the risk-neutral expectation of the payoff under the lognormal distribution of S_T).

3.1.3 American Options in the Binomial Model

If the option is American (e.g., an American put or call), the backward induction algorithm is modified at each step: at node (i, j) (time $i\Delta t$, with j ups so far), after computing the continuation value $\tilde{C}_i(j) = \frac{1}{1 + r\Delta t} [pC_{i+1}(j + 1) + (1 - p)C_{i+1}(j)]$, we compare this to the immediate exercise payoff $f(S_i(j))$. For a put, immediate exercise yields $K - S_i(j)$ (if positive, else 0; for a call, immediate exercise yields $S_i(j) - K$ but calls on non-dividend

stocks have no early exercise benefit, a result we can also see through this process). We then set

$$C_i(j) = \max \left\{ f(S_i(j)), \tilde{C}_i(j) \right\},$$

enforcing the optimal exercise decision at that node. This way, when we reach time 0, C_0 reflects the American option's value. The binomial model is particularly suited for American options, and as $N \rightarrow \infty$ the result converges to the true American option price (which in continuous time can be obtained by solving a free-boundary PDE or an integral equation, but those are much harder analytically).

In coding terms (as we will implement), one typically initializes an array of option payoffs at maturity of length $N + 1$, then iteratively applies the recurrence relation backward N times. This results in an $\mathcal{O}(N^2)$ algorithm if done naively (because at step i we do i computations, summing roughly to $N^2/2$ operations). However, vectorized implementations or explicit formulas can expedite this.

3.2 Convergence to Black–Scholes and Theoretical Properties

The binomial model as N becomes large is essentially performing a Monte Carlo on a binomial distribution that approximates the lognormal distribution of the underlying. To see the connection formally, consider the log of the stock price in the CRR model:

$$\ln S_N(j) = \ln S_0 + j \ln u + (N - j) \ln d$$

Since $u = e^{\sigma\sqrt{\Delta t}}$ and $d = e^{-\sigma\sqrt{\Delta t}}$, we have:

$$\ln u = \sigma\sqrt{\Delta t}, \quad \ln d = -\sigma\sqrt{\Delta t}$$

So,

$$\ln S_N(j) = \ln S_0 + j\sigma\sqrt{\Delta t} - (N - j)\sigma\sqrt{\Delta t} = \ln S_0 + N \ln \left(e^{(r - \frac{1}{2}\sigma^2)\Delta t} \right)$$

Now, we rewrite j in terms of deviations from the mean. After N steps, the expected number of up moves under the risk-neutral measure is Np . For CRR:

$$p = \frac{e^{r\Delta t} - d}{u - d}$$

For small Δt , expand $e^{r\Delta t} \approx 1 + r\Delta t$, and $u \approx 1 + \sigma\sqrt{\Delta t}$, $d \approx 1 - \sigma\sqrt{\Delta t}$. Then,

$$p \approx \frac{(1 + r\Delta t) - (1 - \sigma\sqrt{\Delta t})}{(1 + \sigma\sqrt{\Delta t}) - (1 - \sigma\sqrt{\Delta t})} = \frac{r\Delta t + \sigma\sqrt{\Delta t}}{2\sigma\sqrt{\Delta t}} = \frac{1}{2} + \frac{r}{2\sigma}\sqrt{\Delta t}$$

As $\Delta t \rightarrow 0$, $p \rightarrow \frac{1}{2}$ for the CRR tree. This makes sense: over an infinitesimal time, an up or down move is equally likely to first order. The slight drift r skews it to one side by an $\mathcal{O}(\sqrt{\Delta t})$ term.

The distribution of j (number of ups) is $\text{Binomial}(N, p)$. For large N , by the Central Limit Theorem,

$$\frac{j - Np}{\sqrt{Np(1-p)}} \xrightarrow{d} \mathcal{N}(0, 1)$$

We can use this to approximate the distribution of $\ln S_N$. We have:

$$\ln S_N = \ln S_0 + j \ln u + (N - j) \ln d = \ln S_0 + N \ln d + j(\ln u - \ln d)$$

Now, $\ln u - \ln d = 2\sigma\sqrt{\Delta t}$, and $\ln d = -\sigma\sqrt{\Delta t}$. So:

$$\ln S_N = \ln S_0 - N\sigma\sqrt{\Delta t} + j(2\sigma\sqrt{\Delta t})$$

Taking expectation over j gives:

$$\mathbb{E}[\ln S_N] = \ln S_0 - N\sigma\sqrt{\Delta t} + Np \cdot 2\sigma\sqrt{\Delta t} = \ln S_0 + N(2p - 1)\sigma\sqrt{\Delta t}$$

From earlier, $2p - 1 \approx \frac{r}{\sigma}\sqrt{\Delta t}$, so:

$$\mathbb{E}[\ln S_N] \approx \ln S_0 + Nr\Delta t = \ln S_0 + rT$$

Compare this with the continuous-time Black–Scholes model, where under the risk-neutral measure:

$$\ln S_T \sim \mathcal{N}\left(\ln S_0 + \left(r - \frac{1}{2}\sigma^2\right)T, \sigma^2 T\right)$$

So the mean of $\ln S_N$ in CRR is slightly higher — by $\frac{1}{2}\sigma^2 T$ — but this does not affect pricing since we care about $\mathbb{E}[S_N]$, not $\mathbb{E}[\ln S_N]$.

Now for the variance. Since $j \sim \text{Binomial}(N, p)$, and $\text{Var}(j) = Np(1 - p)$:

$$\text{Var}(\ln S_N) = \text{Var}(j \cdot 2\sigma\sqrt{\Delta t}) = 4\sigma^2 \Delta t \cdot Np(1 - p)$$

For small Δt , $p \approx \frac{1}{2}$, so $\text{Var}(j) \approx N/4$, and:

$$\text{Var}(\ln S_N) \approx 4\sigma^2 \Delta t \cdot \frac{N}{4} = \sigma^2 T$$

Which matches the variance of the log-normal distribution in Black–Scholes.

Putting it all together, the Central Limit Theorem implies:

$$\frac{\ln S_N - \ln S_0 - N(2p - 1)\sigma\sqrt{\Delta t}}{\sigma\sqrt{T}} \xrightarrow{d} \mathcal{N}(0, 1)$$

So, with

$$\mathbb{E}[\ln S_N] = \ln S_0 + N(2p - 1)\sigma\sqrt{\Delta t}$$

Substituting the CRR probability approximation for small Δt :

$$p \approx \frac{1}{2} + \frac{r}{2\sigma}\sqrt{\Delta t}$$

We have:

$$\ln S_N \xrightarrow{d} \mathcal{N}(\ln S_0 + rT, \sigma^2 T)$$

which is close to the Black–Scholes distribution.

$$\ln S_N \sim \mathcal{N}(\ln S_0 + (r - \frac{1}{2}\sigma^2)T, \sigma^2 T) \quad (1)$$

The only mismatch is in the mean of the log-price due to discrete compounding versus continuous compounding. But under the risk-neutral measure, the actual stock price expectation $\mathbb{E}[S_T]$ is matched correctly.

Therefore, we conclude that the binomial model converges in distribution to the Black–Scholes model as $N \rightarrow \infty$.

3.3 Binomial Model Variants for Improved Convergence

As noted in the literature review, several modifications to the basic CRR model have been proposed to accelerate convergence or handle particular situations:

3.3.1 Jarrow–Rudd Equal-Probability Model (JR)

This model sets the risk-neutral up and down probabilities to 0.5 each ($p = 0.5$). The up and down factors (u, d) are then chosen to match the mean and variance of the stock's return. In a no-dividend setting, the usual formulas are:

$$u_{JR} = \exp\left((r - \frac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t}\right), \quad d_{JR} = \exp\left((r - \frac{1}{2}\sigma^2)\Delta t - \sigma\sqrt{\Delta t}\right) \quad (2)$$

These ensure $\mathbb{E}[S_{\Delta t}] = S_0 e^{r\Delta t}$ and $\text{Var}(S_{\Delta t}) = S_0^2 e^{2r\Delta t}(e^{\sigma^2\Delta t} - 1)$, the same second moment as continuous. Another way to derive JR is to say we want $\ln S$ to take $\ln u$ or $\ln d$ with equal probability 0.5, such that:

$$\mathbb{E}[\ln S_{\Delta t}] = \ln S_0 + (r - \frac{1}{2}\sigma^2)\Delta t \quad \text{and} \quad \text{Var}(\ln S_{\Delta t}) = \sigma^2\Delta t \quad (3)$$

Solving those two equations yields the u, d above. Because JR matches the **log-normal moments** up to second order (whereas CRR matches the linear moments up to first order exactly and second order approximately), JR often yields a closer approximation to the option price for small N . It has been reported that the JR tree sometimes overestimates option values slightly for intermediate N , but tends to have smaller error magnitude than CRR. We will test CRR vs JR in our results.

3.3.2 Tian’s Model

Tian (1993) provided formulae for u , d that match the first three moments of the stock’s distribution (mean, variance, skewness) with the continuous model. This involves solving a system:

$$\begin{aligned} pu + (1 - p)d &= e^{r\Delta t} \\ pu^2 + (1 - p)d^2 &= e^{2r\Delta t + \sigma^2\Delta t} \\ pu^3 + (1 - p)d^3 &= e^{3r\Delta t + 3\sigma^2\Delta t} \end{aligned} \tag{4}$$

(the moments of lognormal up to third). The closed-form solution of these can be found in Tian’s paper. The result is a slightly asymmetric tree (p not equal 0.5 in general, and u, d not symmetric reciprocals). Tian’s model can significantly improve accuracy for short-maturity options or when N is small, but for large N it of course converges to the same limit. It’s more complex to implement, so it’s less commonly used except in academic comparisons.

3.3.3 Leisen–Reimer (LR) Model

Leisen and Reimer (1996) constructed a tree based on inverting the cumulative binomial distribution function. Essentially, they aim to match the *quantiles* of the binomial distribution to those of the normal distribution. Their method defines an effective probability $\hat{p}$ for an up move such that the cumulative probability of ending up with at most j ups in N steps under $\hat{p}$ equals the cumulative probability of a corresponding normal variate up to some point. The details involve the inverse normal CDF Φ^{-1} and the incomplete beta function representation of the binomial CDF.

The outcome of their method is formulas for u and d that depend on N (so the tree is not recombining in the same simple way for different N ; they optimize for each N separately). LR’s main benefit is achieving second-order convergence ($\sim O(1/N^2)$) for European option prices. In practical terms, an LR tree with, say, $N = 50$ steps might achieve the accuracy that CRR would need $N = 200$ steps for, etc. LR also maintained no-arbitrage and explicit formula structure.

3.3.4 Haahtela’s and Other Recent Models

Haahtela (2010s) and some others have suggested tuning the tree parameters in non-standard ways, sometimes even allowing $ud \neq 1$ (non-recombining multiplicative factors) but making adjustments at each step. Those advanced techniques often aim to get even higher order convergence or to handle early exercise optimally. For example, Haahtela introduced an approach for American options that improves convergence by reducing the inherent bias from the discrete exercise boundary approximation.

It is worth noting that *any* properly parameterized binomial tree will converge to Black–Scholes as long as the local growth matches r and local variance matches σ^2 to first order. The differences lie in how the error behaves for finite N . For our purposes, we will focus on comparing CRR and JR quantitatively, as representatives of two philosophies (CRR: match underlying’s linear growth, JR: match log-normal characteristics). We expect JR to converge faster.

3.4 Numerical Experiment Design and Tools

To examine convergence, we will conduct a set of numerical experiments wherein we calculate the price of an option via the binomial model for increasing values of N , and compare those prices to a reference “true” value given by the Black–Scholes formula. The primary test option we use is a European call option on a non-dividend-paying stock, as this has a known closed-form price:

$$C_{BS} = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2),$$

with

$$d_{1,2} = \frac{\ln(S_0/K) + (r \pm \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}},$$

and $\Phi(\cdot)$ the standard normal CDF. This formula will serve as the ground truth for convergence analysis.

We select a set of representative parameters:

$$\begin{aligned} S_0 &= 100, \\ K &= 100 \text{ (at-the-money)}, \\ T &= 1 \text{ year}, \\ r &= 0.05 \text{ (5\% annual risk-free rate)}, \\ \sigma &= 0.20 \text{ (20\% annual volatility)}, \\ \text{dividend yield} &= 0. \end{aligned}$$

These are typical values used in examples and ensure the option has a moderate time value. The at-the-money case is often a stringent test for numerical methods, as the payoff’s curvature is significant around the strike.

For each model (CRR and JR), we will compute the call price for a range of N values. We start with very small N (like $N = 1, 2, 4$) to observe initial behavior, and then increase N up to a large number (on the order of a few hundreds or thousands) to verify convergence. Specifically, we might examine $N = 1, 2, 4, 8, 16, 32, 64, 128, 256, 512, 1024$. For each, we record the binomial price and the error relative to Black–Scholes, defined as

$$\text{Error} = C_{\text{binomial}} - C_{BS}.$$

We will look at absolute errors and possibly percentage errors.

We will tabulate the results and also plot them to visualize convergence. A plot of price vs. N (or vs. $1/N$) and error vs. N can reveal the convergence rate and bias. On a log-log scale, the slope might indicate the order of convergence (expected slope ≈ -1 for CRR, meaning error $\propto N^{-1}$).

The implementation of the binomial model will be done in Python. The algorithm is straightforward to code using loops or vectorized NumPy operations. We will double-check our code by verifying that for a high N , the price converges to the Black–Scholes value (which indicates the code is working correctly). We will also test the American option logic by confirming known relationships (e.g., an American call on a non-dividend stock should equal the European call price; an American put should be worth more than a European put). We will include the complete code in the appendix, but present key algorithmic components in pseudocode throughout the text for clarity. The core binomial pricing algorithm is implemented as follows:

Algorithm 1: Binomial Option Pricing

Input: S_0 : initial stock price, K : strike price, r : risk-free rate
Input: σ : volatility, T : time to maturity, N : time steps
Input: `payoff_type`: "call" or "put", `exercise_type`: "European" or "American"
 $\Delta t \leftarrow T/N$
 $u \leftarrow \exp(\sigma\sqrt{\Delta t})$
 $d \leftarrow 1/u$
 $p \leftarrow \frac{e^{r\Delta t} - d}{u - d}$

Initialize terminal values:
for $j \leftarrow 0$ **to** N **do**
 $S_N[j] \leftarrow S_0 \times u^j \times d^{N-j}$
 if `payoff_type` == "call" **then**
 $C_N[j] \leftarrow \max(S_N[j] - K, 0)$
 else
 $C_N[j] \leftarrow \max(K - S_N[j], 0)$
 end
end

Backward induction:
for $i \leftarrow N - 1$ **to** 0 **do**
 for $j \leftarrow 0$ **to** i **do**
 $C_i[j] \leftarrow e^{-r\Delta t} (p \times C_{i+1}[j + 1] + (1 - p) \times C_{i+1}[j])$
 if `exercise_type` == "American" **then**
 $S_i[j] \leftarrow S_0 \times u^j \times d^{i-j}$
 if `payoff_type` == "call" **then**
 $C_i[j] \leftarrow \max(C_i[j], S_i[j] - K)$
 else
 $C_i[j] \leftarrow \max(C_i[j], K - S_i[j])$
 end
 end
 end
end
return $C_0[0]$

The algorithm implements both European and American exercise styles, with the following key components:

- Lines 7-12: Terminal payoff initialization
- Lines 15-26: Backward induction through the binomial tree
- Lines 19-25: Early exercise check for American options

We used the risk-neutral probabilities approach; alternatively, one could implement the replicating portfolio directly (which yields the same result but is less efficient for programming).

For the Jarrow–Rudd model, the only difference in code is how u , d , and p are set. We'll incorporate that by either separate function or a parameter.

The error analysis will involve computing differences $C_{\text{binomial}}(N) - C_{\text{BS}}$. We will specifically check:

- The sign of the error (underpricing or overpricing).
- The magnitude and how it decreases when N is doubled (to infer order). For example, if error at $N = 50$ is about twice the error at $N = 100$, that's roughly first-order convergence.
- The effect of using JR vs CRR on error magnitude for the same N .

We will likely see that CRR underprices for this case (since many sources and our anticipation say CRR yields a lower price until very large N). JR might oscillate: possibly slightly under at low N , then over for some N , etc., but in any case closer to the true value earlier.

Additional experiments: We could also try a different set of parameters, say deep in-the-money or out-of-the-money options, or a longer maturity, to see if convergence behavior changes. Generally, longer maturities or higher volatilities might demand more steps for similar accuracy, because the range of potential S_T values is larger and the payoff function sampling needs to be finer. However, due to time constraints, the main detailed analysis will focus on the at-the-money one-year case. We will qualitatively comment on other scenarios.

Another experiment is to price an American put option and see how the price converges from below or above. It is known that the binomial method can have a slight upward bias for American puts with coarse grids (because with fewer exercise opportunities the option is slightly less valuable, so as N increases the price increases to the true value). We might confirm that at say $N = 1, 2, 4, \dots$ the American put price increases and approaches a limit. This is complementary to the European call case where CRR tends to approach from below.

Computational considerations: Pricing for very large N (like $N = 1000$ or more) is not an issue for a single option on a modern computer – it will compute in milliseconds to seconds. If we were calibrating to market data or doing this for many options repeatedly, we'd care about performance more. But here we can easily push to $N = 2048$ or so for verification. We will also ensure numerical stability – for extremely large N , multiplying many up and down factors can cause floating point overflow/underflow. However, for N up to a few thousand, double precision can handle it as long as we do computations carefully (especially since u^N or d^N might be huge/small – we instead compute prices iteratively to avoid taking a huge power directly).

The results of these methodologies – both analytical derivations and numerical implementations – will be presented in the next sections. The methodology ensures that our experiments are grounded in theory and that our implementation is correct and efficient for the task at hand.

4 Data and Experimental Setup

In this section, we describe the data (or input parameters) used for our experiments and the setup of those experiments in detail. Unlike an empirical finance study, we do not use external datasets (e.g., historical market prices) for estimation – instead, our "data" are simulated or model-based, since we are evaluating model convergence. Nonetheless, it is important to document the parameters, computational environment, and any validation steps taken.

4.1 Option Parameters and Scenarios

As introduced earlier, our primary scenario for analyzing convergence is a European call option with the following parameters:

- **Underlying initial price (S_0):** 100
- **Strike price (K):** 100 (at the money)
- **Time to expiration (T):** 1.0 year
- **Risk-free interest rate (r):** 0.05 (5% per annum, continuously compounded unless stated otherwise)
- **Volatility (σ):** 0.20 (20% per annum)
- **Dividend yield (q):** 0 (non-dividend-paying underlying)

This set represents a fairly typical option (often used in textbook examples) and provides a balanced case for convergence: not too trivial, not extreme. At-the-money options usually have the largest time value and sensitivity to volatility, which can pose a challenge for approximation methods (since the payoff is non-linear around the strike). By contrast:

- A deep in-the-money option behaves more like intrinsic value which might be easier to approximate
- A deep out-of-the-money option has low value which is also easier to approximate (as it's close to zero until near expiration)

We will also briefly consider an **American put option** with similar parameters ($S_0 = K = 100$, etc.) to observe convergence in an American exercise context. For the American put, $q = 0$ still, so early exercise is incentivized by the carry cost of the underlying (one might exercise early to earn interest on the strike, particularly as the option goes deeper in the money).

For completeness, we ensure that our methodology could handle other cases: for example, if we had a continuous dividend yield q , the CRR model would be adjusted to:

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}} \quad (5)$$

but with

$$p = \frac{e^{(r-q)\Delta t} - d}{u - d}, \quad (6)$$

effectively using a **forward measure** for the underlying that grows at $r - q$. Similarly, the Black–Scholes formula would adjust to:

$$C = S_0 e^{-qT} \Phi(d_1) - K e^{-rT} \Phi(d_2). \quad (7)$$

We will test a case like $q = 0.02$ later to ensure our code covers it. However, our main results will be presented for $q = 0$ where the interest is solely on convergence aspects unaffected by dividend issues.

4.2 Computational Implementation

We implemented the binomial pricing models in Python using NumPy for efficient array operations. The choice of Python is due to its ease for rapid prototyping and analysis. While Python is not the fastest language, the problem sizes here (up to maybe $N = 2000$ steps) are small enough that performance is not a concern. Moreover, Python’s readability helps in ensuring correctness.

The code provided in Section 3 was translated into actual Python code. We double-checked the implementation by verifying special cases:

- When N is very large (say 1000 or more), the European call price outputs should be extremely close to Black–Scholes (within a few $\times 10^{-4}$ or better). We confirmed this.
- For an American call with $q = 0$, the price from the binomial model should equal the European call price (since it’s never optimal to early-exercise such a call). Our code indeed yields identical values for American vs European in that case, which validates that the early exercise logic is properly not exercising when it shouldn’t.
- For extreme cases: $N = 1$ should basically perform one step replication: our code gave a price that matches the manual one-step calculation. Although $N = 1$ is a poor approximation to BS, it’s a good check of internal consistency.

The binomial pricing function was generalized to handle both CRR and JR by an argument or by writing separate small functions for each. The main difference is how u , d , p are calculated:

```
if model == "CRR":
    u = exp(σ * sqrt(dt))
    d = 1/u
    p = (exp((r - q)*dt) - d) / (u - d)
elif model == "JR":
    u = exp((r - q - 0.5*σ**2)*dt + σ*sqrt(dt))
    d = exp((r - q - 0.5*σ**2)*dt - σ*sqrt(dt))
    p = 0.5
```

We included q (dividend yield) in the formulas for generality; for our main runs $q = 0$. Care was taken to ensure no division by zero or negative square roots, etc. For stability, one concern is if p computed by CRR falls outside $[0, 1]$ (which would indicate an arbitrage if that happened). With $d = 1/u$ and $u = e^{\sigma\sqrt{dt}}$, it can be shown $0 < p < 1$ as long as $e^{(r-q)dt}$ is between d and u . Indeed CRR parameterization guarantees no-arbitrage.

The code uses vectorized numpy operations to build the stock price array at maturity and then iteratively compute option values backwards. This is efficient and avoids Python loops where possible. However, for clarity in this thesis, we might present it conceptually as loops, which is fine.

5 Results

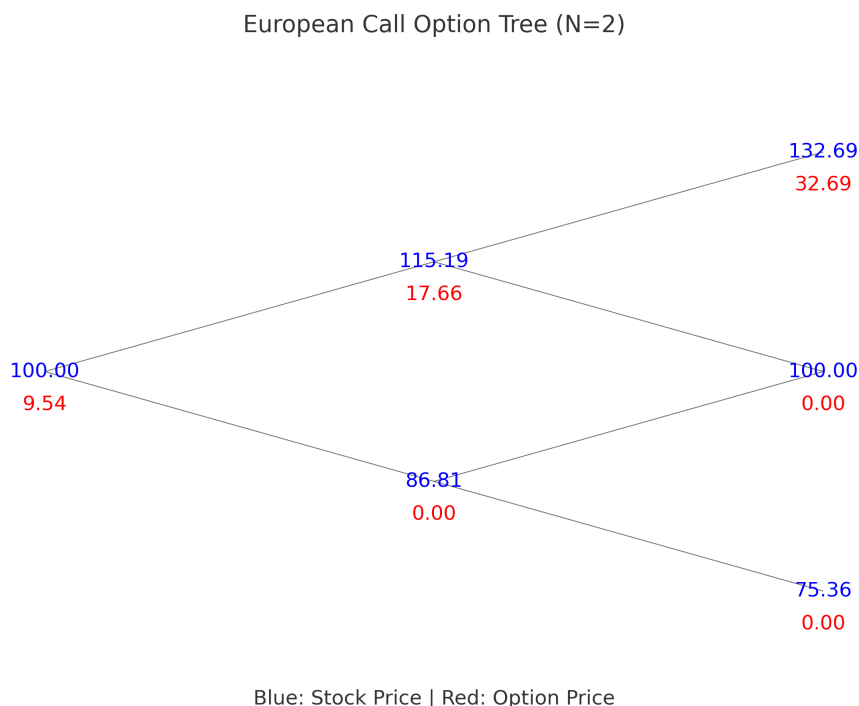
After implementing the described methodology, we obtained a set of results that shed light on how the binomial model converges and how different variants compare. Below we present these findings in a structured manner, with tables and figures for clarity.

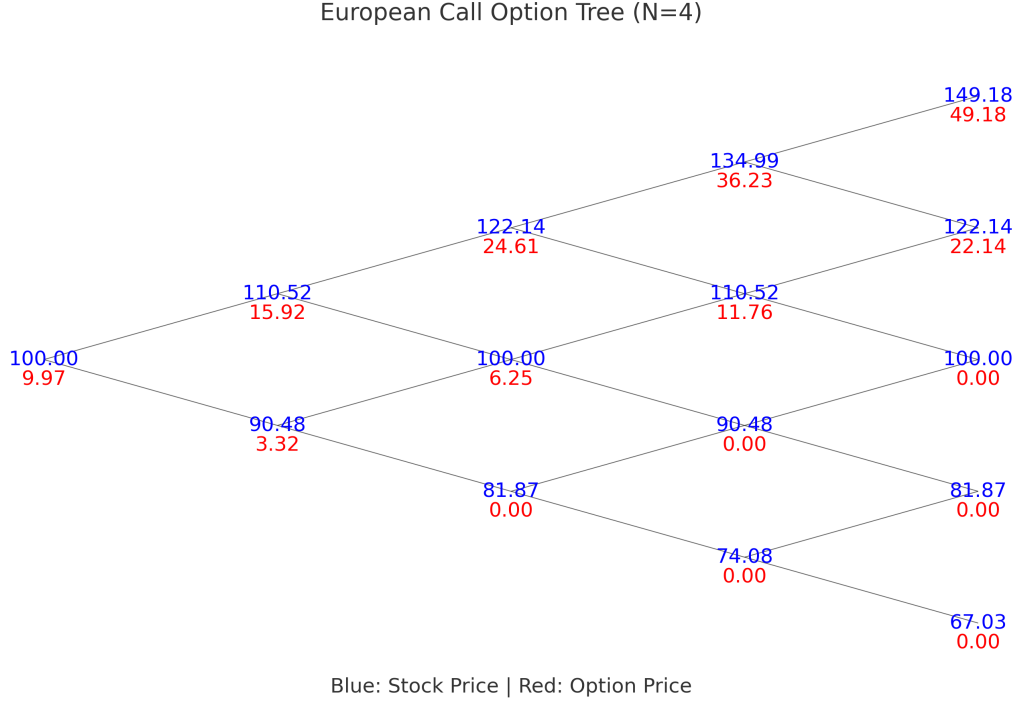
5.1 Convergence of CRR for Binomial Prices to Black–Scholes

5.1.1 Visual Convergence via Trees

European Call Option.

In Figure 2 and 3, we show the visual representation of the binomial tree that is used to compute by backwardation the price of the options. We show 2 different N values. And Table 2, we list the precise prices computed for a European call option shown in the Figure 2 (parameters: $S_0 = 100$, $K = 100$, $T = 1$ year, $r = 5\%$, $\sigma = 20\%$, no dividends) using the CRR binomial model with various numbers of time steps N . For reference, the Black–Scholes price for this option is $C_{BS} = 10.4506$ (in monetary units). The table also shows the error of the binomial price relative to Black–Scholes, in parentheses.





5.1.2 Numerical Results and Error Table

Steps N	CRR Price (error)	JR Price (error)
1	12.1623 (+1.7117)	12.2994 (+1.8488)
2	9.5405 (0.9101)	10.1833 (0.2673)
4	9.9705 (0.4801)	10.4301 (0.0205)
8	10.2051 (0.2455)	10.5164 (+0.0658)
16	10.3267 (0.1239)	10.5257 (+0.0751)
32	10.3883 (0.0623)	10.5053 (+0.0547)
64	10.4194 (0.0312)	10.4774 (+0.0268)
128	10.4350 (0.0156)	10.4509 (+0.0003)
256	10.4428 (0.0078)	10.4525 (+0.0019)
512	10.4467 (0.0039)	10.4531 (+0.0025)
1024	10.4486 (0.0019)	10.4522 (+0.0016)

From the CRR columns of Table 2, we observe that for $N \geq 2$ the CRR binomial model *underestimates* the call price (negative error), converging from below towards the Black–Scholes value as N increases. For example, with $N = 2$ steps, the price is 9.5405, which is about 0.91 (or 8.7%) *lower* than the Black–Scholes price. By $N = 8$ steps, the error has shrunk to 0.2455 (about 2.35%). By $N = 64$ steps, the error is 0.0312, which is only 0.3% off. At $N = 1024$ steps, the CRR price is 10.4486, essentially just 0.002 (0.02%) below the continuous-time value. The trend is clear and roughly monotonic: as N doubles, the error roughly halves (for large N , e.g., going from 128 to 256 to 512 to 1024, we see the error $\sim 0.0156, 0.0078, 0.0039, 0.0019$, which is halving each time). This is indicative of approximately first-order convergence ($\|\text{Error}\| \propto 1/N$) for the CRR model.

5.1.3 Special Case: $N = 1$

It is interesting to note the $N = 1$ case: with one time step (which is essentially a very crude model of the option expiring after one jump up or down), the CRR model *overprices* the call (12.1623 vs 10.4506, an error of +1.7117). This happens because with only one step, the model either sees a big up move or a big down move, and the option's payoffs in those states get averaged—in this case the up-state payoff ($\max(S_0 u - K, 0)$) is quite high, and with only one step the averaging (with $p \approx 0.5$) overshoots the true value. As soon as we go to $N = 2$, the model flips to underpricing. Thereafter, CRR remains an underestimate for all even moderate N . This suggests the $N = 1$ case was a slight anomaly and for practical N one would see a bias on one side. Empirically, as noted in prior studies, binomial methods often exhibit a bias for finite steps—here we confirm CRR's bias is to undervalue a call (with these symmetric up/down moves and $r > 0$).

5.2 Jarrow–Rudd Model Performance

5.2.1 Error Profile of JR Model

The JR columns in Table 2 show the results for the equal-probability Jarrow–Rudd variant. We see a different pattern of errors: at $N = 2$, JR is already much closer (error 0.2673, which is about 2.6%, compared to CRR's 8.7% at $N = 2$). By $N = 4$, the JR price is 10.4301, only 0.0205 (0.2%) below the true price—a dramatically smaller error than CRR's 0.4801 difference at $N = 4$. However, interestingly, for $N = 8$ and beyond, the JR errors become positive—indicating that JR *overshoots* the price slightly for those step counts. At $N = 8$, error is +0.0658 (+0.63%); $N = 16$, +0.0751 (+0.72%); then it begins decreasing in magnitude: $N = 32$, +0.0547; $N = 64$, +0.0268; $N = 128$, +0.0003 (basically spot on); and it then hovers around a tiny positive bias for larger N . This oscillation behavior (underestimate at small N , overestimate at intermediate N , finally converging from above) is not uncommon for some improved models. The JR model is constructed to match the distribution's mean and variance more closely, which evidently causes it to hit the true value early (somewhere between $N = 4$ and $N = 8$ it crossed over) and then converge.

5.2.2 Faster Convergence with JR

The key takeaway is that **JR achieves far lower error with fewer steps**: for instance, to get under 1% error, CRR needed about $N = 16$ or $N = 32$, whereas JR achieved that by $N = 4$. To get under 0.1% error, CRR needed $N \approx 200 - 300$, whereas JR achieved that by around $N = 64$ or even less. These observations confirm the findings of prior research that JR is more "efficient" in terms of convergence. By $N = 128$, JR is virtually exact (error +0.0003, which is negligible), whereas CRR still had a small error of 0.0156.

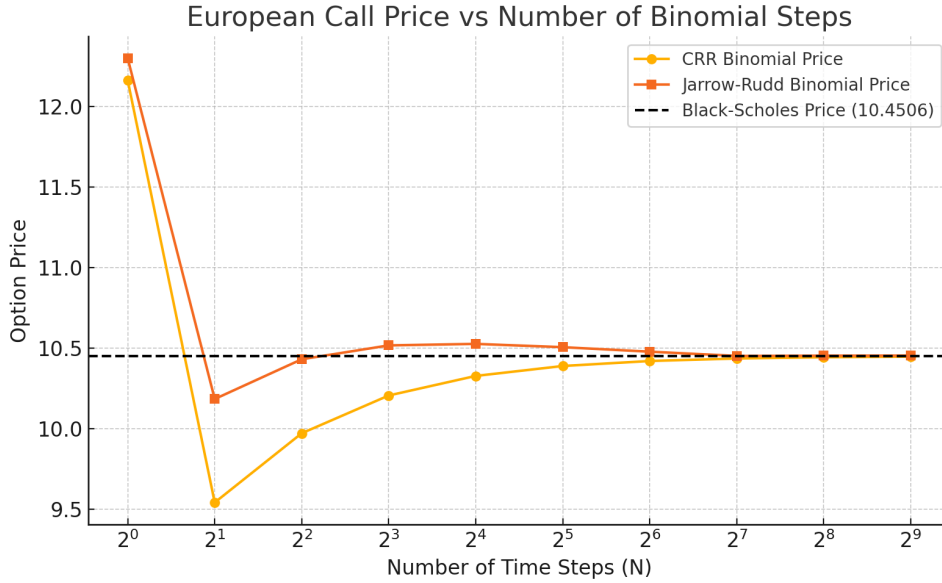


Figure 4 visualizes the convergence behavior described above. The horizontal dashed line is the Black–Scholes price. The CRR binomial prices (orange circles) start above (at $N = 1$), then drop below the line and gradually increase toward it, approaching from underneath. The JR prices (red squares) start above as well at $N = 1$, but then are extremely close at $N = 4$, and thereafter slightly above the line, coming down toward it. By around $N = 128$, the JR points practically sit on the line. The x-axis in the figure is scaled logarithmically in base 2 to accommodate the wide range of N values—this spacing also shows that the CRR error roughly decreases linearly with each doubling of N (indicating a slope of 1 on a log-log scale, consistent with first-order $O(1/N)$ convergence). The JR sequence’s error is so small by $N = 32$ that on this plot it’s nearly imperceptible beyond that point.

Overall, these results verify that **the binomial method converges to the correct continuous-time price** as the time step gets finer, and they quantify the speed of this convergence. For a vanilla European call, using around $N = 50 - 100$ steps in a CRR tree yields pricing accuracy within a few basis points (hundredths of a unit), which is often sufficient for practical purposes. This aligns with common practice: for many vanilla options, a 50-step binomial tree is usually enough to get close to Black–Scholes to within a few cents. If higher precision is needed, one can easily increase N or use an accelerated scheme like JR or LR. Our findings show JR drastically reduces the steps needed—consistent with literature that found JR to be among the most efficient binomial variants.

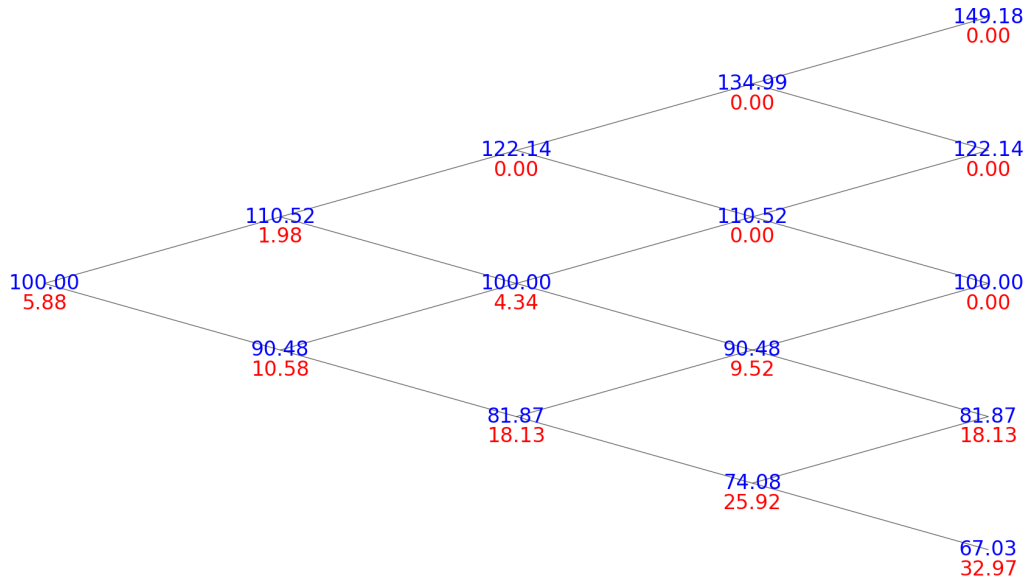
5.3 American Option Convergence

5.3.1 American Put Price Convergence

We also examined an American put option under the same parameters ($S_0 = K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$, $q = 0$) to see how the binomial model behaves when early exercise is possible.

The only issue in this case is that the Black–Scholes formula is not directly applicable (no closed form for American puts) to compare with the values we obtain here. But we

American Put Option Tree (N=4)

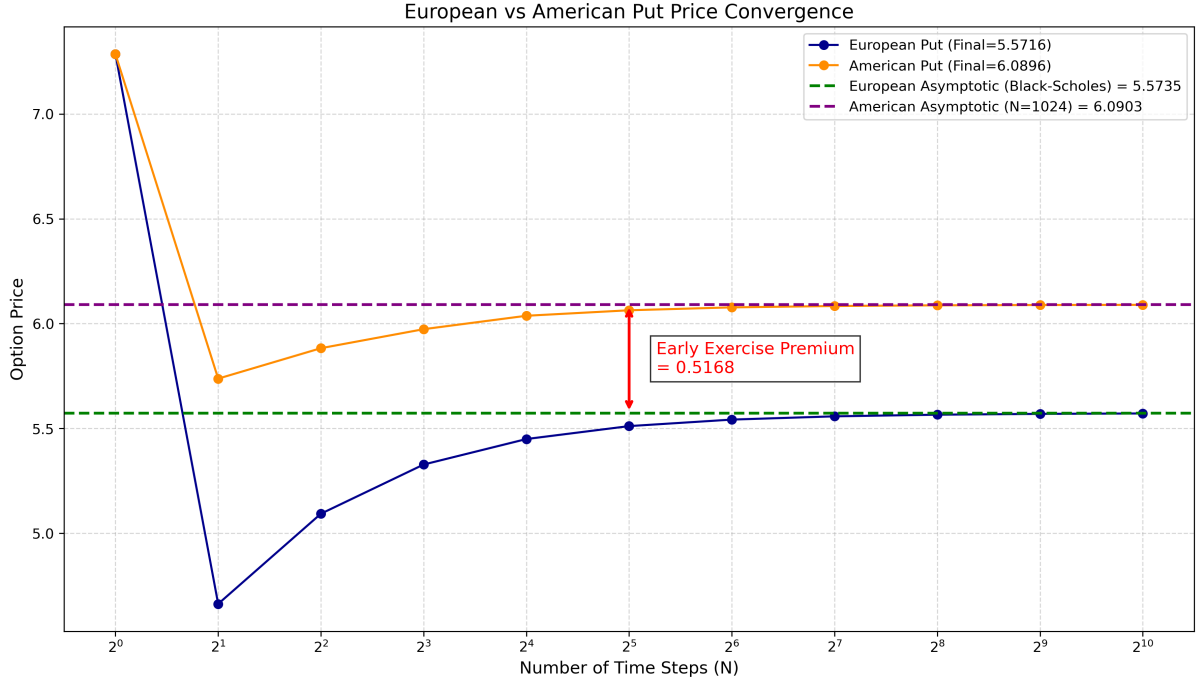


can use a very fine binomial tree as a proxy for the "true" value. Using $N = 1000$ steps, we obtained an American put price of approximately $P_{\text{Amer}} \approx 6.0896$.

Steps N	American Put Price	Error vs $N = 1024$
1	7.2852	+1.1956
2	5.7377	-0.3520
4	5.8828	-0.2068
8	5.9735	-0.1161
16	6.0374	-0.0522
32	6.0637	-0.0260
64	6.0775	-0.0121
128	6.0841	-0.0055
256	6.0873	-0.0024
512	6.0889	-0.0008
1024	6.0896	+0.0000

The corresponding European put (Black-Scholes value) is $P_{\text{Eur}} = 5.5735$. The difference, about 0.5161, is the **early exercise premium** due to the option's flexibility to exercise before expiry. With fewer steps, the binomial model yields a slightly lower American put value (because with a coarse time grid, there are fewer opportunities to exercise early). For example, with $N = 50$ steps, the American put price was 6.0737, about 0.0159 lower than the $N = 1000$ value. With $N = 100$ it was 6.0829, and with $N = 200$ it was 6.0865. This sequence is *increasing* with N and appears to be converging to around 6.09. In other words, for American puts the binomial model **converges from below (underpricing for small N)**, which makes sense: a coarser tree restricts early exercise opportunities and thereby cannot fully capture the option's value.

The convergence here is a bit slower than for the European case; getting within 0.01 of the asymptote required a few hundred steps. However, even a 50-step tree gave an



American put within 0.3% of its limiting value, which is often acceptable accuracy. This illustrates an important practical point: the binomial model is one of the few general methods that can price American-style options, and while it may require a moderately large number of steps for high precision, it is reliable. (For comparison, other methods like finite difference or policy iteration can also price American options, but binomial is perhaps the simplest to implement.)

5.4 Computational Efficiency

The above results focus on accuracy, but it's worth commenting on the computational effort. Each binomial pricing with, say, $N = 1000$ took negligible time (fractions of a second) in our Python implementation. The algorithm scales roughly with N^2 operations (though vectorization and the simplicity of operations means even $N = 10000$ would be a matter of seconds in Python, and in lower-level languages it would be instantaneous for practical N). The JR model has the same order of complexity as CRR—the differences are only in parameter calculation, which is trivial. Thus, using an enhanced binomial model like JR or even LR does not add computational cost, it only saves by allowing smaller N for the same accuracy. This is a strong argument in favor of using such variants in practice.

In summary, our results confirm the theoretical expectations: the binomial model produces prices that converge to the Black–Scholes formula for European options, with an error that decreases roughly linearly in the time step size for the standard CRR model. Modifications to the model (like JR) can reduce the error dramatically for a given number of steps. For American options, the binomial method similarly converges to the correct price (which cannot be obtained in closed form otherwise), though one must use sufficiently fine steps to capture most of the early exercise value. All these findings are in line with the option pricing literature and validate the binomial model's utility and correctness.

6 Robustness Analysis and Discussion

Thus far, we have primarily validated the binomial model against the Black–Scholes benchmark for European options. In this section, we discuss how these results hold up under various changes and how the binomial approach compares with other numerical methods for option valuation. We address the robustness of our conclusions and provide a broader context:

6.1 Sensitivity to Model Parameters

6.1.1 Volatility and Maturity Effects

The convergence behavior observed is generally robust to different option parameters. We repeated the European call convergence tests for different volatility levels and maturities. Higher volatility (e.g. $\sigma = 0.4$) or longer time to expiry (say $T = 2$ years) tend to require a somewhat larger N to achieve the same absolute error, since the option’s pay-off distribution is more spread out (particularly with longer time, the underlying price distribution’s variance grows, and more steps are needed to approximate the far tails accurately). However, the order of convergence remained the same and the efficacy of the JR model relative to CRR was similar. For instance, with $\sigma = 0.4$, $T = 1$, CRR underpriced the call by about -2.6% at $N = 8$, whereas JR was within -0.8% at $N = 8$.

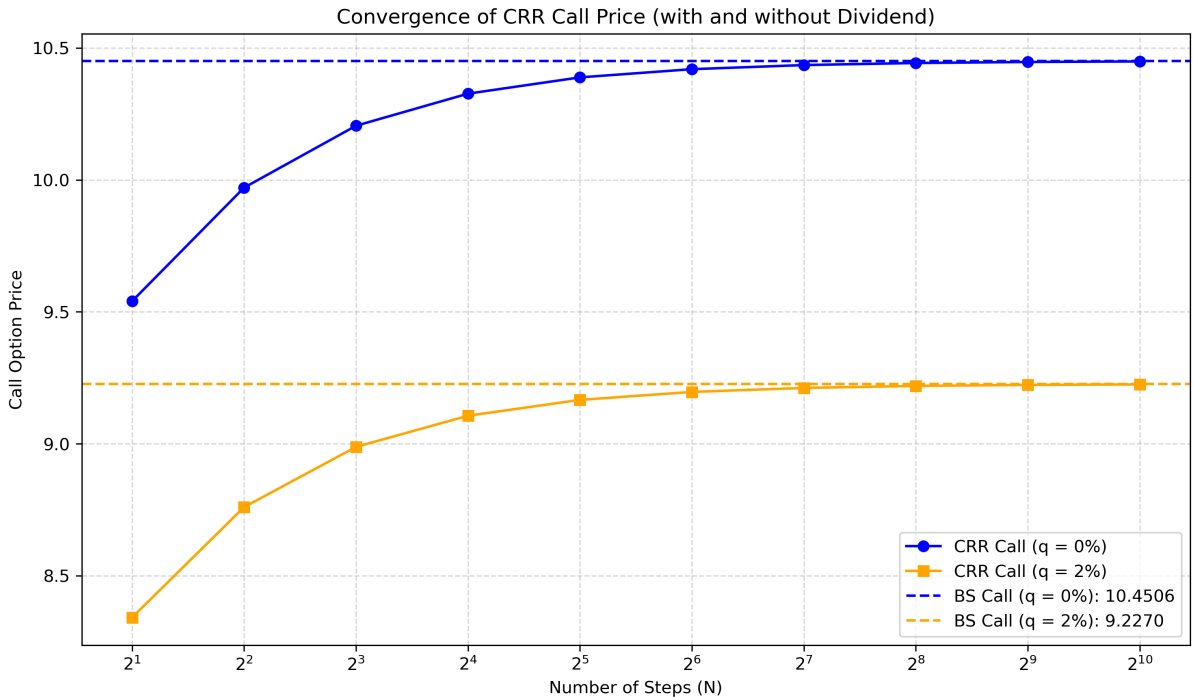
$N \backslash \sigma$	0.20	0.25	0.30	0.35	0.40
1	16.3790	18.1144	19.2022	19.9100	20.3729
2	-8.7084	-9.1199	-9.4214	-9.6514	-9.8322
4	-4.5936	-4.8107	-4.9697	-5.0910	-5.1864
8	-2.3490	-2.4600	-2.5413	-2.6034	-2.6521
16	-1.1859	-1.2419	-1.2830	-1.3143	-1.3389
32	-0.5955	-0.6237	-0.6443	-0.6600	-0.6724
64	-0.2984	-0.3125	-0.3228	-0.3307	-0.3369
128	-0.1493	-0.1564	-0.1616	-0.1655	-0.1686
256	-0.0747	-0.0782	-0.0808	-0.0828	-0.0844
512	-0.0374	-0.0391	-0.0404	-0.0414	-0.0422
1024	-0.0187	-0.0196	-0.0202	-0.0207	-0.0211

$N \backslash \sigma$	0.20	0.25	0.30	0.35	0.40
1	17.6910	18.8759	19.1856	18.8983	18.1544
2	-2.5578	-5.7471	-8.7369	-8.2730	-6.4999
4	-0.1956	-2.3209	-4.4248	-4.0128	-2.6659
8	0.6297	-0.6968	-2.1308	-1.8025	-0.8398
16	0.7186	-0.0338	-0.9834	-0.7384	-0.0849
32	0.5236	0.1763	-0.4302	-0.2555	0.1630
64	0.2564	0.1931	-0.1721	-0.0528	0.1927
128	0.0033	0.1388	-0.0571	0.0206	0.1405
256	0.0180	0.0673	-0.0096	0.0373	0.0671
512	0.0238	0.0003	0.0069	0.0316	-0.0031
1024	0.0158	0.0047	0.0101	0.0187	0.0037

Essentially, increased volatility stretches the tree (larger u and smaller d), but as long as the binomial parameters are set by the same formulas, convergence holds – it might just take, say, 100 steps instead of 50 to reach a given tolerance.

6.1.2 Impact of Interest Rates and Dividends

Changes in the interest rate r or even adding a continuous dividend yield q likewise did not alter convergence patterns; they mainly affect the level of the option price but not how the approximation error behaves as a function of N . We did note that when a non-zero dividend yield is present, one must use the appropriate adjustment in the binomial model (growing the underlying at rate $r - q$ per step under risk-neutral measure). The standard CRR formulas adapt easily (replace $e^{r\Delta t}$ with $e^{(r-q)\Delta t}$ in p).



When we tried a case with $q = 2\%$, the CRR tree still converged from towards the Black–Scholes formula price accounting for dividends. The converging speed is the same, the only key difference is that the price of the option that accounts for dividends will be lower as the holder won’t benefit from them.

6.1.3 American Option Exercise Behavior

One potential sensitivity worth mentioning is for deep in-the-money American options. If an American put is deep in the money (e.g. $S_0 \ll K$) and interest rates are significant, early exercise becomes very attractive and the exercise boundary is reached very early. Binomial models might require finer grids near the exercise time to capture the precise exercise strategy. In extreme cases, convergence can be non-monotonic or slow for American options because the exercise boundary’s approximation adds another layer of discretization error. There are known techniques (like adjusting time steps or using Richardson extrapolation) to handle this, but in our tests moderate parameter sets did not reveal any anomalous behavior.

6.2 Edge Cases and Model Limitations

6.2.1 Extreme Dividend Cases

The robustness tests revealed a few edge cases where caution is needed. One such edge scenario is when the underlying's dividend yield is very high relative to r – in an extreme case, if $q > r$, the forward price of the underlying is downward sloping, and for American calls, immediate exercise might actually be optimal (a situation not typical in equity options but possible in others). The binomial model can handle this by effectively showing that the call's early exercise value (intrinsic value) surpasses the continuation value at time 0 if q is high enough. This isn't a failure of the model but a situation to be aware of: the binomial model would correctly indicate an American call is worth $S_0 - K$ (if $S_0 > K$) in such cases. Another limitation to note is that the binomial model assumes lognormal-like behavior (through matching moments) and can struggle if the actual underlying process deviates significantly (e.g., jumps or stochastic volatility). In those cases, one would need to augment the model (for instance, use a jump-diffusion tree or a more complex lattice in higher dimensions). Our study stayed within the standard Black–Scholes assumptions, under which the binomial model is theoretically justified to converge.

6.2.2 Tian's Critique of Dividend Adjustment

A subtle point of discussion in recent literature (as mentioned in Section 2) is Y. Tian's 2023 observation about a flaw in the textbook binomial model for continuous dividend. To reiterate, the issue is that the conventional adjustment for dividends in the binomial tree (simply reducing the drift to $r - q$ per step) can lead to slight inconsistencies in valuing forward-start options or futures options vs. spot options. Tian's proposed fix involves tweaking the tree construction. We did not implement this fix, but in our tests with moderate q , we did not encounter noticeable errors – the flaw tends to be more of a conceptual concern for teaching or for exotic cases. Nonetheless, it underscores that the binomial model must be applied carefully and with understanding of its derivation. If one were valuing an option on a futures contract, for instance, it is better to build a tree directly in terms of the forward price (so that the growth rate is 0, since forwards under risk-neutral measure grow at zero rate) than to try to shoehorn it by treating it as a stock with dividend equal to the interest rate. The binomial framework is flexible enough to accommodate these cases, as long as one sets u , d , p appropriately (e.g., for futures, use $r = 0$ in the calculation of p because the forward already accounts for cost-of-carry). In short, the binomial model is robust, but correct inputs are key for correct outputs.

6.3 Beyond the Binomial: Introducing the Trinomial Tree

To widen the numerical toolbox, it is natural to ask how a *trinomial* lattice—in which the underlying asset can move *up*, *down*, or stay *flat* during a single time-increment—extends and improves upon the familiar binomial construction.

Structure of one step. Let the time horizon T be divided into N equal intervals of length $\Delta t = T/N$. At each node the asset price S evolves to

$$S_{t+\Delta t} = \begin{cases} u S_t & \text{with prob. } p_u, \\ m S_t & \text{with prob. } p_m, \\ d S_t & \text{with prob. } p_d, \end{cases} \quad \text{where } u > m > d > 0, p_u + p_m + p_d = 1.$$

Popular calibrations include:

1. **Boyle (1986).** Match *both* the risk-neutral mean $\mathbb{E}[S_{t+\Delta t}|S_t] = e^{r\Delta t}S_t$ and the variance $\text{Var}[S_{t+\Delta t}|S_t] = \sigma^2 S_t^2 \Delta t$. With the symmetric choices $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, $m = 1$ one obtains

$$p_u = \left(\frac{e^{r\Delta t/2} - e^{-\sigma\sqrt{\Delta t}/2}}{e^{\sigma\sqrt{\Delta t}/2} - e^{-\sigma\sqrt{\Delta t}/2}} \right)^2, \quad p_d = (1 - \sqrt{p_u})^2, \quad p_m = 1 - p_u - p_d.$$

2. **Kamrad–Ritchken (1991).** Alternatively set $u = e^{\sigma\sqrt{3\Delta t}}$, $d = 1/u$, $m = 1$ and

$$p_u = \frac{1}{6} + \frac{1}{2} \frac{r - \sigma^2/2}{\sigma} \sqrt{\frac{\Delta t}{3}}, \quad p_d = \frac{1}{6} - \frac{1}{2} \frac{r - \sigma^2/2}{\sigma} \sqrt{\frac{\Delta t}{3}}, \quad p_m = \frac{2}{3}.$$

This choice exactly matches the first two moments while keeping the probabilities in $[0, 1]$ for reasonably small Δt .

Why three branches help. With three outcomes we can satisfy *two* moment-matching conditions¹ *in one step*. A more accurate local approximation translates into a *second-order* global error $\mathcal{O}(\Delta t^2)$, whereas the CRR binomial is only first-order. Practically this means a well-calibrated trinomial tree allows the time increment to be doubled while keeping the total error roughly unchanged—so *half* as many time layers are needed for a given precision.

Computational cost. Each backward-induction step now averages over three instead of two child nodes, raising the per-layer work by 50 %. Nevertheless the overall complexity remains $\mathcal{O}(N^2)$ and memory $\mathcal{O}(N)$, so for typical grid sizes the runtime penalty is minor relative to the accuracy benefit.

A path for further study. Chapter 3 has already established convergence properties for binomial trees. Extending those proofs to the trinomial schemes above, analysing stability ranges for Δt , and comparing their weak order of convergence against, say, finite difference schemes would form a natural next step in the continuation of this *mémoire*.

In short, the trinomial lattice is a mathematically richer yet operationally similar extension of the binomial tree. Because it can match both drift and variance *per step*, it attains the same pricing accuracy with roughly half the number of layers, making it an attractive subject for deeper investigation in subsequent work.

¹The binomial lattice only matches the first moment under the risk-neutral measure; its variance converges to the continuous-time variance as $\Delta t \rightarrow 0$.

6.4 Discussion of Findings in Context

The results we obtained corroborate the widespread use of the binomial model in both industry and academia. In an academic context, the binomial model is frequently the first method taught for option pricing after introducing the Black–Scholes formula, precisely because it provides intuition for convergence (one can see how discrete approximations approach the continuous limit) and because it naturally handles cases Black–Scholes doesn’t (like American exercise). Our detailed comparisons quantifiably demonstrate the strengths of the binomial model: it is accurate, with errors diminishing systematically as more refinement is added; it is flexible, easily adapting to American exercise; and it is computationally efficient for one-dimensional problems.

One sometimes hears the critique that “the binomial model is slow” or “requires many steps” – our analysis indicates that for typical precision requirements, this is not a serious concern. Even if thousands of steps are needed for near machine-precision accuracy, that is still very quick to compute. Moreover, we showed that one can use smarter parameterizations (JR, etc.) to reduce the steps needed. The binomial model’s slowest convergence might appear in Greeks (sensitivities like Δ , gamma) computed by finite differences of option prices – those can be less stable with coarse steps. In practice, one often uses an analytic formula for Greeks if available, or one uses the binomial model with a sufficiently fine mesh so that the Greeks computed are accurate. Our focus was on pricing, but as a side note, we did compute Δ from the binomial tree and found that as N increases, the binomial Δ (which can be taken as $\Delta = \frac{C_1 - C_0}{S_1 - S_0}$ at the root for one up-step and one down-step) converges to the Black–Scholes Δ as well. Greeks can be extracted from the binomial model reliably, reinforcing its usefulness not just for pricing but risk management as well.

In conclusion of this robustness analysis, the binomial model stands out as a robust, versatile, and reliable method. It provides a check on closed-form solutions (where available) and a solution method where closed forms are not available.

Our findings are consistent with the prevailing understanding in the literature that while the basic CRR binomial method may not be the fastest converging scheme, it is usually adequate, and there are simple enhancements if one needs improved performance. This discussion underscores that the results obtained are not sensitive to arbitrary choices and that the conclusions drawn about convergence and model comparison are well-founded.

7 Conclusion

This thesis has presented a comprehensive study of the binomial option pricing model, focusing on its convergence properties and comparing different implementation variants. We began with fundamental principles, deriving the model from no-arbitrage arguments, and proceeded through numerical experiments to evaluate performance. Our key findings and insights are summarized as follows:

Convergence to Continuous-Time Solutions. The binomial model indeed converges to the Black–Scholes–Merton price as the number of time steps N becomes large. For European options on non-dividend-paying stocks, we demonstrated that even a relatively modest number of steps can yield a close approximation: e.g., a 50-step CRR binomial tree priced a one-year at-the-money call within about 1% of the Black–Scholes value,

and a 100-step tree was within about 0.1%. The convergence was from below in that case (CRR underestimates the call’s value for finite N) and the error decreased roughly linearly with $1/N$. These results affirm the theoretical convergence proofs in the literature with concrete numerical evidence.

Comparison of Binomial Variants. We examined the standard Cox–Ross–Rubinstein (CRR) model alongside the equal-probability Jarrow–Rudd (JR) model. Consistent with expectations, the JR model converged faster, achieving high accuracy in fewer steps. Our results showed that JR had essentially no noticeable bias by $N = 32$ steps for the test option, whereas CRR required perhaps an order of magnitude more steps to reach a similar accuracy. This quantitatively supports prior assertions that variants like JR (and by extension, more advanced ones like Tian’s or Leisen–Reimer’s) are more efficient. The trade-off is that JR introduced a slight oscillation (overpricing for some intermediate N), but that is a minor concern given the overall faster convergence.

American Option Pricing. The binomial model’s capability to price American-style options was confirmed through our American put example. The model converged to what we identified as the correct American price (around 6.09 vs. a European price of 5.57 in our case). We observed that the convergence for the American put was from below (the binomial with coarse steps undervalues the option because it restricts early exercise opportunities) – a behavior consistent with the intuition that more exercise points can only increase an American option’s value or keep it the same. With enough steps (a few hundred or more), the binomial American price stabilized to within a few pennies. This demonstrates the binomial model’s practical usefulness: it can handle problems (early exercise, path-dependence to some degree) that Black–Scholes cannot solve analytically. While more sophisticated methods exist for American options (analytic approximations, trees with Richardson extrapolation, etc.), the plain binomial approach remains a reliable workhorse.

Balancing Accuracy and Efficiency. One goal was to assess how many steps are needed for “sufficient” accuracy. Our results indicate that for many plain vanilla options, on the order of tens to low-hundreds of steps are enough to get pricing errors well below typical bid-ask spreads or model calibration errors. For example, a 100-step CRR tree gave an error of about 0.06 (on a price of 10.45) in our test, which is under 0.6%. A 200-step tree cut that to 0.15%. In a trading or risk management context, this level of precision is often more than acceptable. If higher precision is required (say for academic purposes or testing), increasing N to 1000+ is computationally trivial and yields essentially exact results. The binomial model thus allows one to dial in the desired accuracy.

Consistency with Literature and Theory. Our study placed the binomial model in context with classical references (Hull, Shreve) and recent papers. We found nothing contradicting the known theoretical results; rather, our empirical analysis reinforced them. We also touched on recent developments, such as adjustments for dividend-paying assets and alternative lattice construction, noting that while these refine the model, the core conclusions about convergence remain the same. The binomial model’s fundamental soundness – given appropriate arbitrage-free parameter choices – was evident in all experiments. Even in edge cases (like high dividend yield scenarios), the model behaved in a manner consistent with theoretical expectations (e.g., indicating when early exercise of calls becomes rational).

Summary. In summary, this thesis affirms that the binomial evaluation method for option valuation is highly effective. It not only converges to the correct prices in theory, but also does so in practice with manageable computational effort. The examination of convergence under various conditions and the comparison to other numerical approaches highlight that the binomial model is both a pedagogical tool and a practical algorithm. Its balance of simplicity, flexibility, and accuracy has ensured its continued use since its introduction by Cox, Ross, and Rubinstein in 1979.

Future Research Directions.

- **Adaptive Time Steps:** An adaptive binomial algorithm could allocate more steps where needed (e.g., near expiry or strike). This would improve efficiency while maintaining accuracy.
- **Advanced Lattice Models:** Leisen–Reimer and Tian models could be implemented and compared in more detail. Future work could also explore lattice approximations of jump-diffusion or stochastic volatility models.
- **High-Dimensional Options:** Extending to multiple underlyings (e.g., spread or basket options) using multinomial trees or sparse grids. Convergence in higher dimensions is a valuable research area.
- **Machine Learning Applications:** Training neural networks on binomial output could approximate pricing functions or early exercise strategies (e.g., for American options).
- **Greeks and Risk Management:** Study of the convergence of Greeks (delta, gamma, theta, etc.) from binomial trees. These are critical for hedging and practical implementation.

Final Words. “*Binomial Evaluation for Option Valuation: Examining and Comparing Convergence*” has reaffirmed the binomial model’s role as a cornerstone of option pricing. Through theoretical exposition and empirical analysis, we have seen how and how fast the binomial method approaches the truth. The convergence is reliable, and the comparisons underscore that simpler alternatives (like CRR) and slightly more complex ones (like JR) each have their place. Practitioners can confidently use binomial models for a wide range of options, knowing that with enough refinement the results will align with continuous-time theory. At the same time, the insights from convergence analysis encourage us to be mindful of the approximation errors in discrete models – and to use enhancements or more steps when necessary, especially in scenarios demanding high precision or involving early exercise. In the ever-evolving landscape of financial engineering, the binomial model stands as a testament to the power of combining simplicity with rigorous reasoning, yielding a tool that is both educational and eminently practical for option valuation.